

Design and Implementation of UAV Authentication System based on

Proof-of-History Blockchain Technology

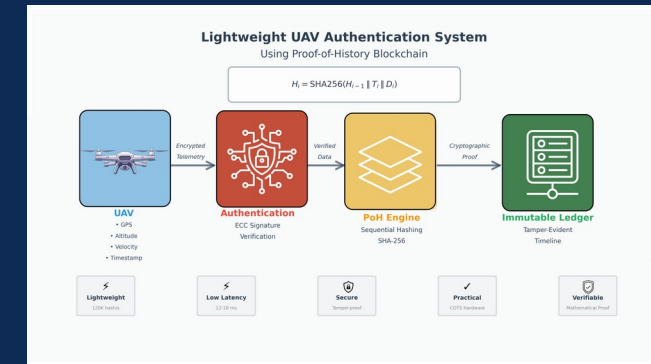
Student: MUNTASIR AL MAMUN

Student No.: F22040119

Major: Computer Science and Technology

Institution: NJUPT — School of Computer Science

Year: 2026



Presentation Outline

1 Introduction & Motivation

3 Related Work & Research Gap

5 PoH Engine Design

7 Implementation Details

9 Evaluation – Security Tests

2 Research Objectives & Contributions

4 System Architecture

6 Authentication Protocol (ECC)

8 Evaluation – Performance

10 Conclusion & Future Work

Introduction & Motivation

Why do UAVs need a new security paradigm?

UAVs are transforming logistics, agriculture, surveillance & emergency response

Critical Security Threats:

- Spoofing & command hijacking over wireless links
- Unauthorized telemetry manipulation & replay attacks
- Centralized architectures = single point of failure
- Forged timestamps → undetectable event reordering

Why existing solutions fall short:

- Traditional PKI: no tamper-evident logging
- PoW / PoS blockchains: too heavy for UAV hardware
- Symmetric keys: scale poorly in multi-UAV fleets

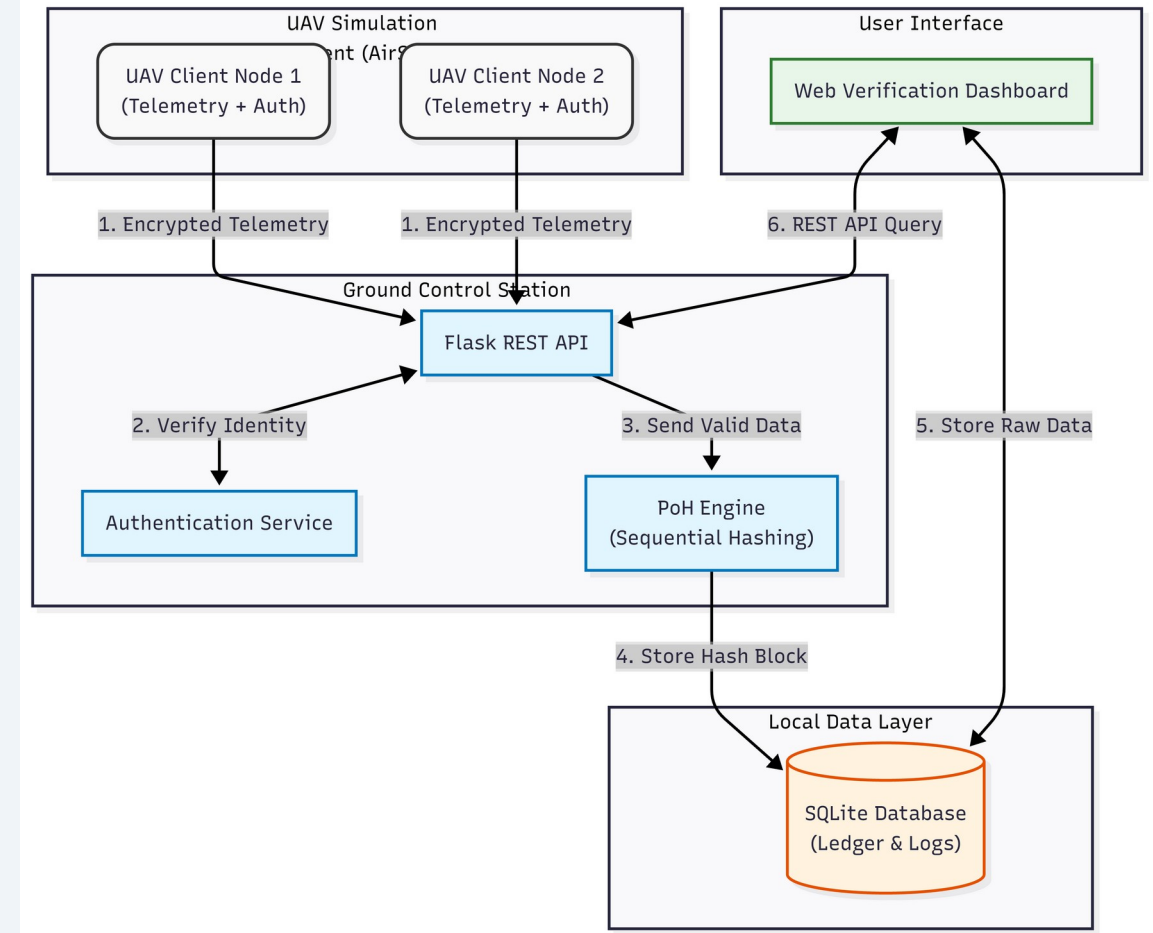


Fig. 1 – UAV Data Integrity Challenges & PoH Motivation

Research Objectives & Contributions

Objectives

1. Develop a PoH-based sequential hashing engine for UAV telemetry
2. Integrate lightweight ECC mutual authentication
3. Implement in AirSim + PX4 SITL simulation environment
4. Design real-time web verification dashboard
5. Evaluate performance & security against common attack vectors

Key Contributions

- ✓ First PoH-based UAV verification framework bridging blockchain principles with real-time flight data
- ✓ Full end-to-end prototype: authentication + sequential logging + verification dashboard
- ✓ Empirical performance & security evaluation under multi-UAV AirSim/PX4 conditions
- ✓ Open-source modular architecture as a foundation for distributed UAV blockchain research

Related Work & Research Gap

UAV Authentication

ECC-based mutual auth, 5G-AKA, RF fingerprinting — lack tamper-evident logging

Blockchain for UAVs

PoW/PoS/DPoS: too heavy. Permissioned chains lack inherent temporal ordering

PoH & VDF Research

Strong sequential-time proofs but no integrated lightweight UAV solution

Hybrid Logging

High-res local logs + periodic anchors — but no unified auth-integrity linkage

Research Gap: No existing system unifies PoH temporal ordering with lightweight ECC authentication for real-time, resource-constrained UAV environments.

System Architecture

Modular service-oriented framework

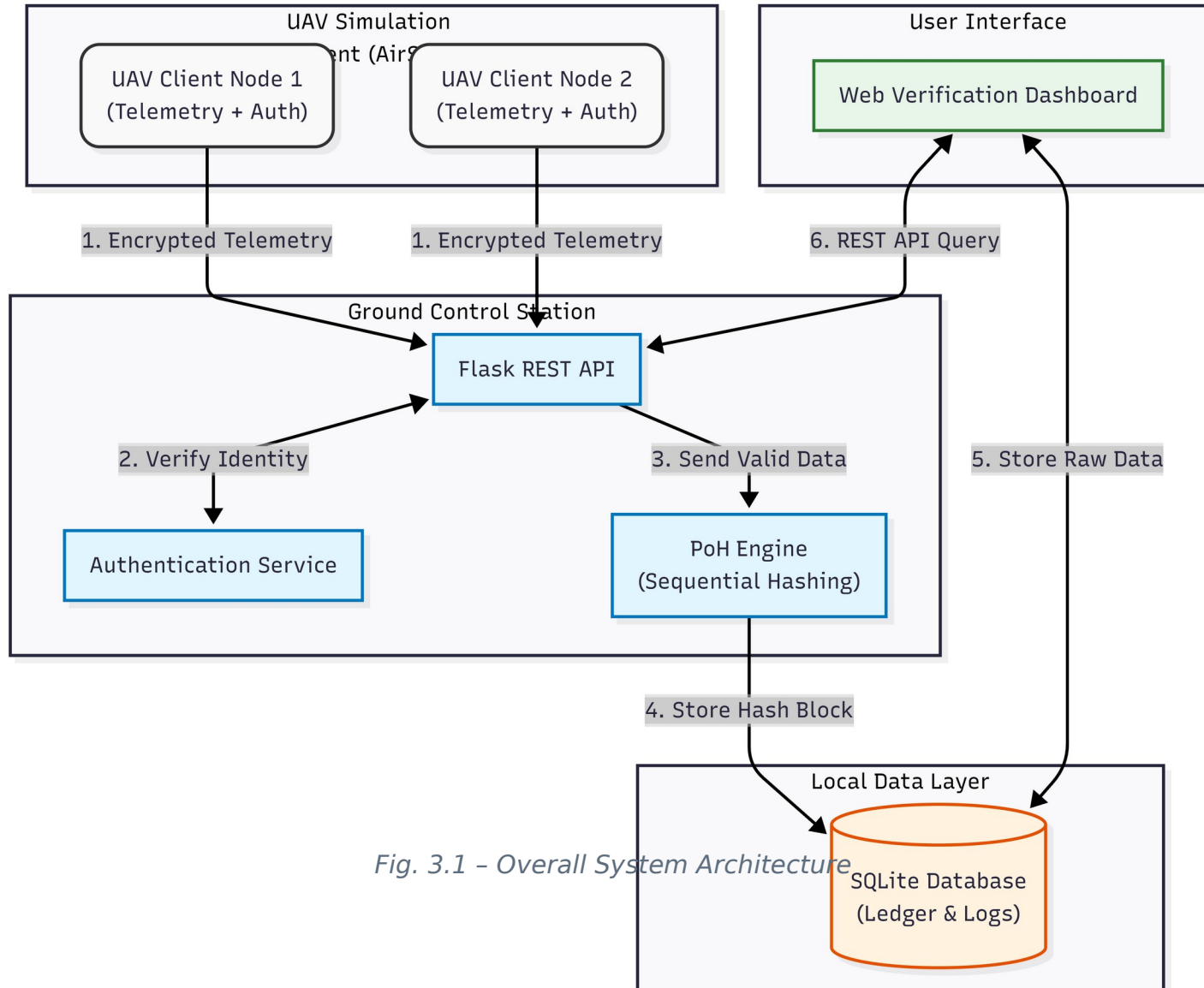


Fig. 3.1 - Overall System Architecture

UAV Client Nodes

AirSim-simulated drones; autonomous flight + telemetry broadcasting via MAVLink

Ground Control Station

Authentication hub + PoH ledger node; Flask REST API; ECC key registry

PoH Blockchain Engine

Python SHA-256 sequential hashing; tamper-evident event timeline

Web Dashboard

Real-time telemetry viz; block verification; forensic log export

Proof-of-History Engine

Cryptographic backbone — tamper-evident sequential timeline

Core Formula:

$$H_i = \text{SHA256}(H_{i-1} \parallel \text{Event}_i \parallel \text{Timestamp}_i)$$

Key Properties:

- Inherently sequential — cannot be parallelised
- Any block alteration invalidates all later hashes
- Verification is embarrassingly parallel (fast audit)
- No distributed consensus needed

Performance:

- > 120,000 hashes/sec on i5-11400H
- Block creation < 1 ms per event
- Full-chain verification: 180–260 ms

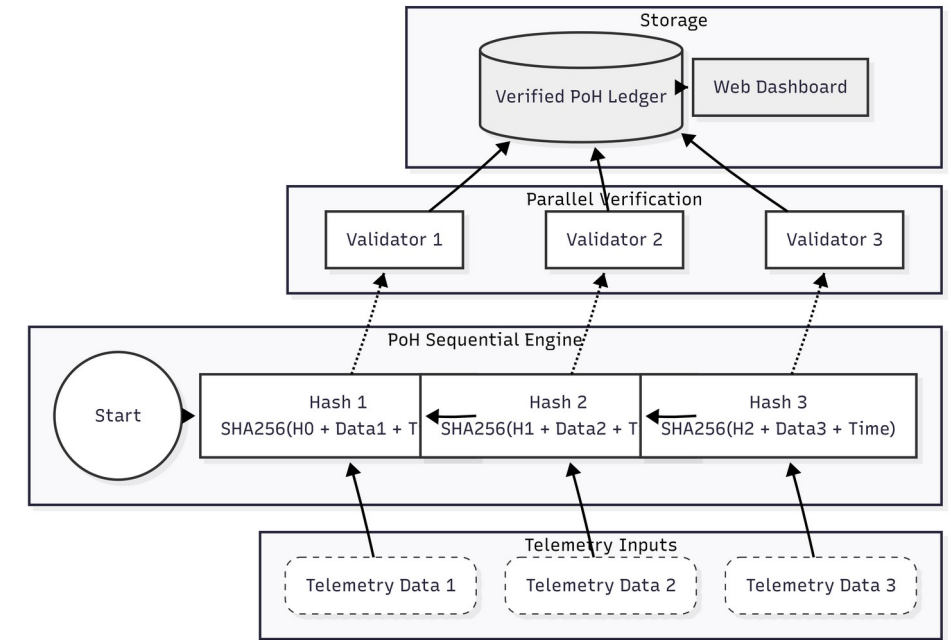


Fig. 3.2 – PoH Hash Sequence & Verification

Authentication Protocol

ECC-based lightweight mutual authentication

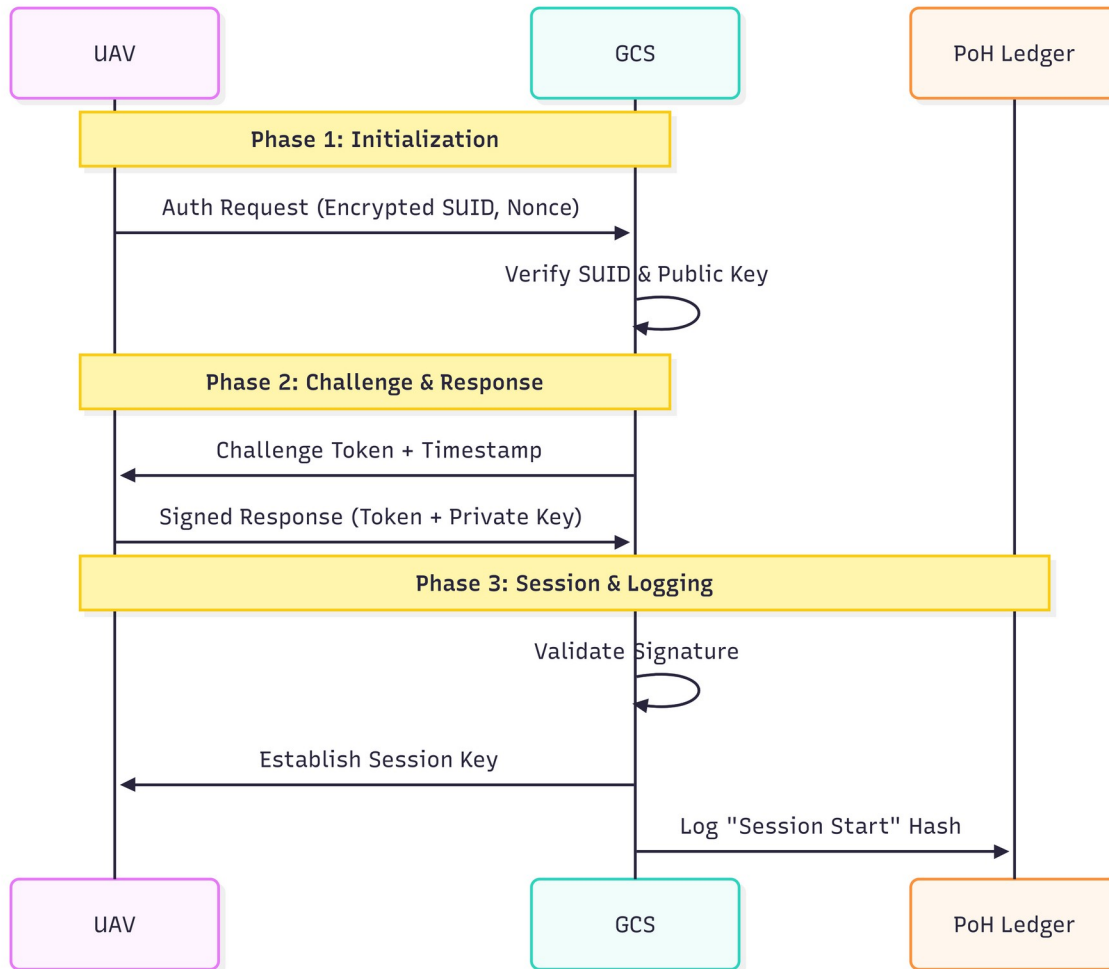


Fig. 3.3 – UAV-GCS Three-Phase Authentication Flow

Why ECC?

Equivalent security to RSA with smaller key sizes → ideal for resource-constrained UAV hardware

3-Phase Handshake:

Phase 1 - Initialization:

UAV sends encrypted temp ID + signed nonce

Phase 2 - Challenge:

GCS verifies signature, issues random challenge token

Phase 3 - Verification:

UAV signs challenge; GCS verifies → session key K_{Session}

Session Security:

- K_{Session} derived via one-way KDF
- Each packet carries MAC(message, K_{Session})
- Session initiation event embedded in PoH chain
- Prevents spoofing, replay & command injection

Data Model & Block Format

Three-layer verifiable ledger structure

Layer 1 - Transaction Data

UAV_ID, Timestamp, GPS, Altitude, Velocity, System Status, Challenge tokens & Session Keys (KTx)

Layer 2 - PoH Metadata

Block Index (i), Previous Hash (H_{i-1}), Current Hash (H_i), Timestamp (T_i), GCS Node Signature (Sig_Node)

Layer 3 - Verification Record

Links session keys to PoH index → retrospective session verification without re-running authentication protocol

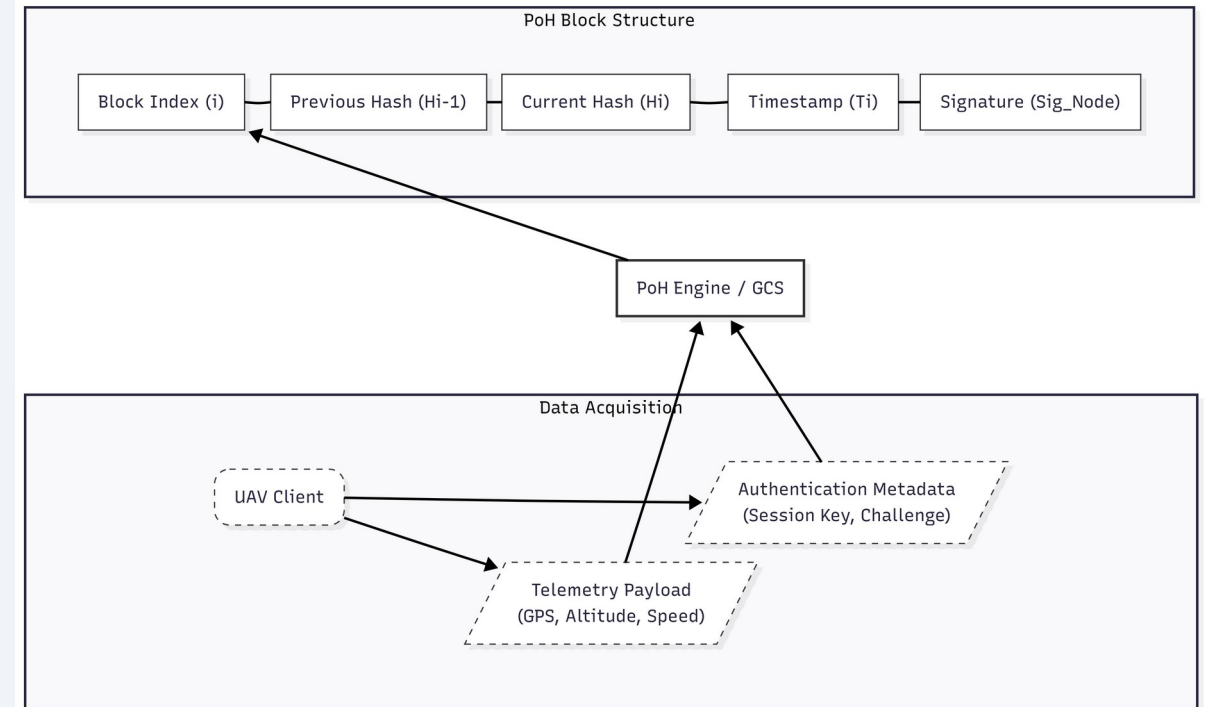


Fig. 3.4 - PoH Ledger Block Structure

Implementation Details

Technology stack & simulation environment

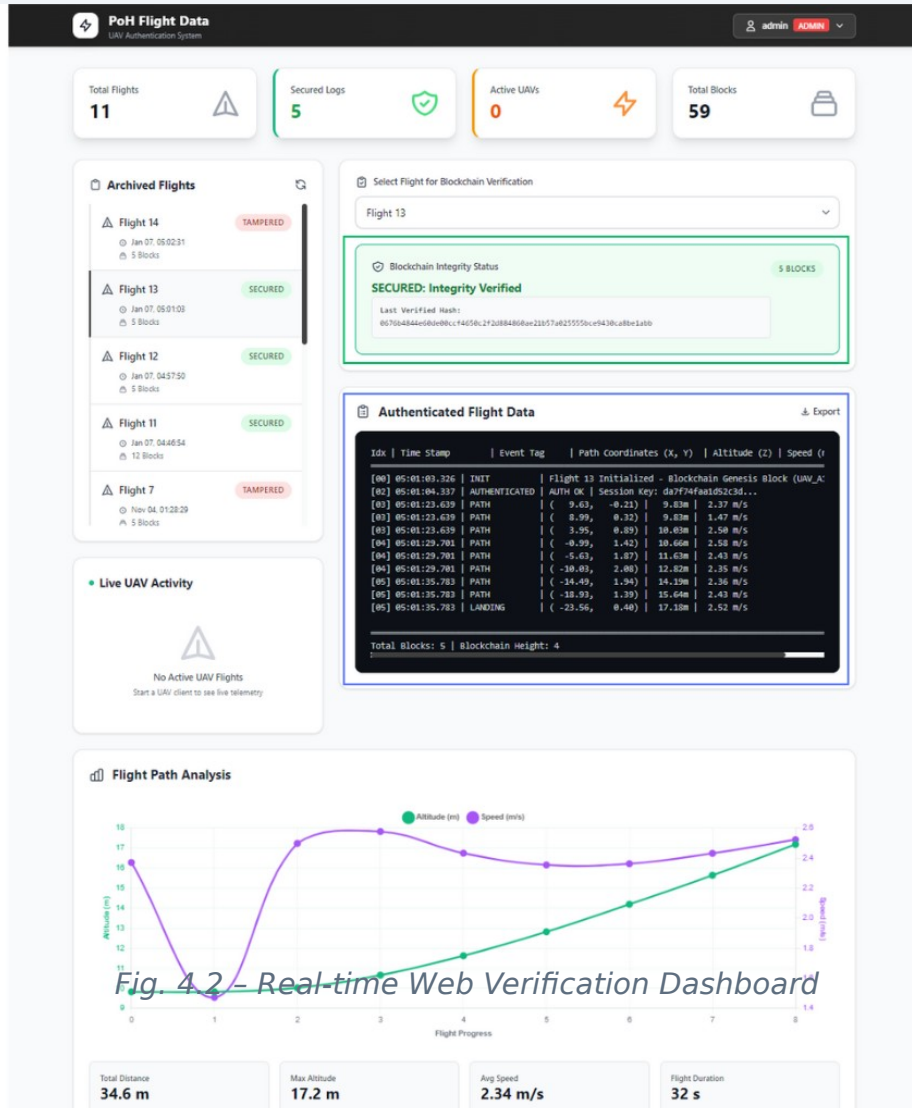


Fig. 4.2 – Real-time Web Verification Dashboard

Hardware Platform:

Intel i5-11400H @ 2.7 GHz | 16 GB DDR4 | NVIDIA GTX 1660 Ti |
Windows 11 / WSL Ubuntu 22.04

Software Stack:

Python 3.11 + Flask 3.0 · PyCryptodome (ECC/SHA-256)
SQLite 3.41 + SQLAlchemy ORM · Jinja2 + AJAX dashboard
Microsoft AirSim + PX4 SITL · MAVLink protocol

API Endpoints:

/api/telemetry — receives signed UAV packets → PoH chain
/api/verify — recomputes hashes, returns integrity token
/api/logs — authenticated historical blockchain access

Dashboard Panels:

Telemetry Viz · Blockchain Ledger · Authentication Log · Export
(JSON/CSV)

Evaluation – Performance Results

AirSim + PX4 SITL multi-UAV simulation

Approach	Auth Latency	Telemetry Log	Ordering	UAV Suitability
This Work (PoH)	~ms (12-18 ms)	< 1 ms/block	Deterministic	★★★★★
PKI-based Auth	Low-Medium	Not supported	None	★★★☆☆
PoW Blockchain	High	Sec-Min	Probabilistic	★☆☆☆☆
PoS Blockchain	Med-High	Seconds	Probabilistic	★★☆☆☆
DAG Ledger	Medium	Sub-second	Partial	★★★☆☆

Key Metrics: Auth cycle 12-18 ms · Telemetry ingestion 6-10 ms · Hash throughput 110-125 K/s · Block latency < 3 ms · Full-chain verify 180-260 ms

Evaluation – Security Tests

Four adversarial scenarios verified

✗ Impersonation Attack

Spoofed SUPI / invalid SUCI → authentication rejected immediately; no PoH entry created

✗ Replay Attack

Captured packets have stale timestamps misaligned with advancing PoH sequence → classified stale & rejected

✗ Data Tampering

Modified block → hash mismatch detected; entire downstream chain flagged invalid (see dashboard alert)

✗ Malicious Chain Fork

Attacker cannot regenerate valid sequential hashes at comparable speed → forked chain rejected by GCS

✓ All four attack categories successfully mitigated by the PoH + ECC framework

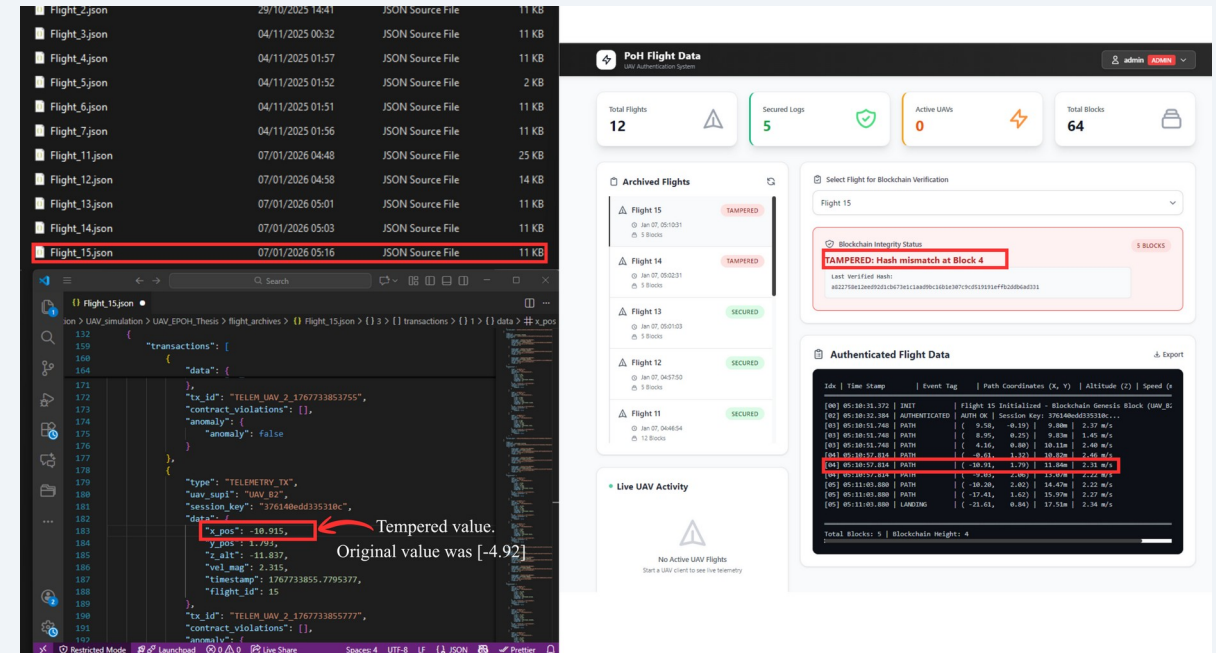


Fig. 5.1 – Data Tampering Attack & PoH Detection Response

Conclusion & Future Work

Summary

- ✓ Novel PoH-based UAV authentication framework — verifiable tamper-evident event timeline without heavy consensus
- ✓ ECC mutual authentication binds verified UAV identities directly to the PoH ledger
- ✓ Real-time web dashboard enables live monitoring and forensic audit
- ✓ Strong resilience against impersonation, replay, tampering & chain-fork attacks
- ✓ Practical performance: ms-level latency, 110-125K hashes/sec throughput

Future Work

Multi-node Distributed PoH

Replicate ledger across GCS nodes for fault-tolerance & scalability

UAV Swarm & Inter-UAV Auth

Handover procedures, cross-UAV verification in BVLOS missions

Physical Hardware Deployment

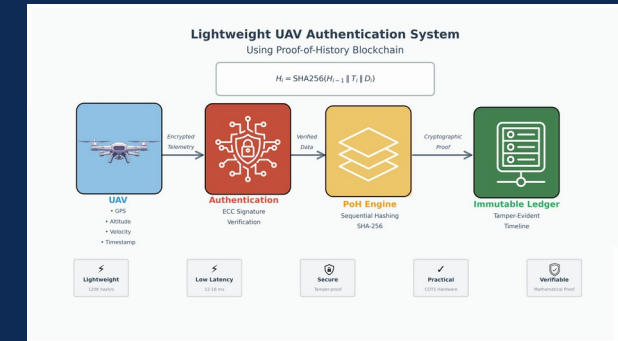
Real-world energy, latency & communication constraints on embedded controllers

Cloud Integration

Periodic checkpoint anchoring to cloud ledger for regulatory compliance

Thank You

Questions & Discussion



Design and Implementation of UAV Authentication System based on Proof-of-History Blockchain Technology

MUNTASIR AL MAMUN

Student No.: F22040119 | Computer Science & Technology
Nanjing University of Posts and Telecommunications | 2026